

Designing Human-in-the-Loop for Mobile Intelligent Notification Systems: Exploring User Preferences for Timing and Form

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Abstract

Human-in-the-loop (HITL) mechanisms offer a promising approach to personalizing intelligent smartphone notification systems, yet little is known about how users prefer to interact with these systems in practice. To address this gap, we conducted semi-structured interviews with 33 participants, using a series of design probes to elicit user reflections on when and how they are willing to express notification preferences, as well as their attitudes and concerns toward incorporating HITL into notification systems. Our findings reveal that users prefer to engage in HITL interactions as part of their existing notification-handling routines, adjusting their willingness to participate based on their attentional and physical resources. While users generally favored lightweight, low-effort interactions, they were not only more willing but also prefer to invest additional time and effort when doing so enabled them to more precisely express preferences that could better support the system's tasks.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Human-in-the-Loop, Intelligent Notification System, Personalization, Preference, Timing

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1 Introduction

Prior research has proposed a range of intelligent notification systems aimed at reducing interruptions and helping users manage and process incoming notifications. These include systems such as *NotiManager* [15], which supports automatic sorting and categorization; *NotiSummary* [26], which generates summaries of notification content; *DataHalo* [8], which visualizes notification importance through ambient visual cues; and *NotiSpeculate* [2], which encourages users to reflect on the potential content of notifications based on alert characteristics.

While these systems assist users in interacting with notifications, most collect user preferences only during the initial setup—typically via onboarding questionnaires or configuration settings—and offer limited support for the ongoing refinement of those preferences over time. This presents a critical limitation, as user preferences are often difficult to fully articulate at the outset and tend to evolve with changes in context, goals, and life circumstances [3, 9, 14, 18]. To enhance personalization, prior work has explored the use of human-in-the-loop (HITL) mechanisms [1, 6, 20, 29], which enable systems to elicit user preferences through either explicit input or implicit behavioral signals. HITL approaches have also been widely adopted in content-based recommendation systems, where lightweight interactions—such as clicking, liking or disliking, labeling, or rejecting content—allow the system to incrementally learn from user feedback and adapt its behavior over time [10, 11, 13].

Yet, despite the growing recognition of the need for personalization in notification systems [2, 8, 9, 14, 15, 18, 19], it remains unclear what forms of HITL interaction are most acceptable, effective, and engaging to users in this domain. Notifications differ markedly in both function and meaning—ranging from reminders and task prompts to updates, alerts, and news [3, 14, 15]. They also vary across dimensions such as time sensitivity, importance, attractiveness, and actionability [7, 14, 15, 23–25, 28].

Furthermore, notification systems may support a diverse set of processing tasks, including filtering [18], alert adjustment [2, 12, 16], device selection [19], categorization [15], deferral [21, 22, 27], summarization [26], and attention guidance [8]. These tasks differ in

input-output structure, interaction modality, and presentation format, and may thus require distinct forms of user input and engagement. As a result, there may not be a one-size-fits-all HITL solutions for notification systems—either from a technical perspective or in terms of user receptiveness.

In addition, prior research suggests that users' willingness to engage in system-driven tasks is influenced not only by the nature of the task but also by the timing of its delivery, as different tasks impose varying cognitive and attentional demands [5]. This raises further questions about how timing may shape receptivity to HITL interactions in notification contexts. Given these complexities, there is a clear need to investigate not only what forms of HITL interaction users find acceptable, but also when such interactions are most likely to be welcomed and acted upon.

Because user participation is critical to the success of HITL-driven personalization, it is essential to understand how users perceive, accept, and engage with different HITL strategies—both in terms of interaction form and timing. To address this gap, this study investigates the following research questions:

- **RQ1:** What forms of human-in-the-loop interaction are preferred by smartphone users in the context of notification systems?
- **RQ2:** What timings of human-in-the-loop interaction are preferred by smartphone users in the context of notification systems?

2 The Interview Study

2.1 Recruitment and Participant

We adopted a multi-phase recruitment strategy to ensure diversity in notification behaviors. In the initial phase, we recruited ten participants through word of mouth and advertisements posted on local social networking platforms. These advertisements included a link to our screening questionnaire that gathered demographic information and notification-related metrics. These metrics included the approximate number of notifications present in participants' notification drawers and the types of notifications they typically received. This phase aimed to ensure variation in both demographic backgrounds and notification interaction patterns.

Subsequently, we adopted a theoretical sampling approach [4, 17], in which recruitment, interviewing, and analysis occurred in parallel. Specifically, based on preliminary insights from the initial data analysis, we recruited ten additional participants in the second wave, focusing on deepening emerging themes and broadening representativeness in notification practices. Recruitment continued until we reached theoretical saturation [4], the point at which no new concepts or insights emerged from additional interviews.

In total, we recruited 33 participants, comprising 11 students and 22 non-students, with 18 identifying as female and 15 as male. Participants ranged in age from 20 to 64. Each participant received a stipend of NT\$300 (approximately US\$9.15) as compensation for their time.

2.2 Design Probes for Interview

To explore how participants imagined expressing notification preferences in real-world contexts, we developed a set of design probes (see examples in Figure 1) to serve as prompts during the interviews.

These probes helped participants focus on concrete interaction scenarios and articulate their expectations and mental models of HITL for notification systems. Rather than narrowing the discussion, the probes acted as provocations to stimulate reflection and elicit descriptions of notification preferences and usage patterns. The variety of these design probes allowed us to examine how different HITL implementations may influence users' perceptions, cognitive load, and willingness to engage.

2.3 Interview Procedure

All 33 participants were interviewed in face-to-face sessions. Each session was attended by two or more researchers: one serving as the moderator, the other(s) observing and taking notes. The moderator provided a consent form and informed participants about the research objective and interview topics. Before beginning the interview, the moderator reviewed the consent form, confirmed participants' agreement, and ensured their responses would be recorded and fully transcribed.

Each interview began with warm-up questions covering two areas: notification-related experiences and system learning experiences. The notification-related questions explored notification types, habits for attending to and acting upon notifications, and notification preferences. The system learning questions explored participants' familiarity with personalization mechanisms, based on their experiences using recommendation systems, social media feeds, and AI tools such as ChatGPT. This phase was intended to gather early insights and help participants become comfortable with the interview topics.

To allow flexibility in exploring emerging themes, we used a semi-structured interview format. The moderator asked participants about their preferred methods for providing notification preferences to a system. For each preference, we followed up to understand their reasoning. We explored variations by modifying notification types, interactions, and contexts to refine their ideal interaction scenarios. Design probes were introduced only when participants had difficulty generating ideas.

2.4 Data Analysis

We adopted a grounded approach [4] to analyze the interview data, with data collection and analysis occurring concurrently. The analysis was supported by the qualitative coding software ATLAS.ti¹ and involved the iterative development of a codebook to guide the interpretation process.

In the initial phase, the research team reviewed interview notes and transcripts from the first ten participants to generate a preliminary list of codes. Each code included a definition to ensure consistent understanding among team members. To establish consistency and clarify any ambiguities, the first three transcripts were coded jointly by all researchers. These sessions helped refine code interpretations and build inter-coder agreement.

Subsequent transcripts were coded independently by different researchers. During this phase, the team held regular meetings to reflect on and revise the codebook. When new codes emerged or existing ones needed clarification, the changes were discussed and implemented only after team consensus. This iterative coding

¹<https://atlasti.com>.

Sorting

Do you want **messaging** notifications to be at the top?

System inference: You tend to prioritize handling messaging notifications.

☐ Yes
☐ No

Please drag and reorder the notifications to your preferred order.

The system's current confidence score is 3 out of 10.

Manager 4 hours ago
Mention this in tomorrow's meeting

Mom 2 hours ago
Are you coming home for dinner tonight?

Professor 4 mins ago
Check this paper.

abc_123456 1 hour ago

What characteristics should a notification have to be placed at the top?

☐ Task-related
☐ Sent by supervisor
☐ Financial
☐ Mentions critical times
☐ Requires interaction with others
☐ Others []

How should notifications from the **Weather** category be handled across a day?

Positioned at the top ☐ Morning Afternoon Evening
Does not affect sorting ☐
Positioned at the bottom ☐

Categorization

The system marks this notification as **Urgent**. Which reason explains this best?

David 2 hours ago
When's the meeting tomorrow?

☐ It requires interacting with others
☐ It asks *When*
☐ It mentions work-related keywords

What keywords make this notification placed in the **Work** category?

Manager 4 minutes ago
Mention this in tomorrow's meeting

☐ Manager
☐ 4 minutes ago
☐ Tomorrow
☐ Meeting
☐ Others []

Determine the ideal categories for this notification.

The system currently classifies this notification as **Important, Work, and Messaging**.

Professor 4 minutes ago
Check this paper.

☒ Important
☒ Work
☐ Family
☐ Advertisement
☐ Urgent

Should notifications like this be in the **To-Do** category?

The system notices you often treat these as pending tasks.

Professor 4 minutes ago
Check this paper.

☐ Yes
☐ No

Summarizing

The system is unsure how to summarize this notification.

Which summaries of this notification would you prefer?

abc_123456 1 hour ago
This restaurant seems great!

☐ abc_123456 shared a restaurant
☐ abc_123456 wants to chat
☐ Someone on Instagram wants to chat
☐ abc_123456 sent a chat message an hour ago

Which notifications need to be summarized?

☐ Manager 4 minutes ago
Mention this in tomorrow's meeting

☐ John Doe Yesterday
Project Documents ...

☐ abc_123456 1 hour ago
This restaurant seems great!

☐ Professor 4 minutes ago
Check this paper.

What elements of this notification should be in the summary?

The model considers this as **non-work messaging**, so it includes its sender and topic in the summary.

Manager 1 hour ago
Send me the proposal tomorrow

☐ Source App (e.g., Slack)
☐ Sender (e.g., Manager)
☐ Received time (e.g., 1 hour ago)
☐ Topic (e.g., proposal)
☐ Tone (e.g., serious)
☐ Deadline (e.g., tomorrow)

Edit the system generated summary for these three notifications if unsatisfied.

David 2 hours ago
When's the meeting tomorrow?

Manager 4 minutes ago
Mention this in tomorrow's meeting

John Doe Yesterday
Project Documents ...

David is asking about the meeting time tomorrow. The manager wants to mention the discussed topic in tomorrow's meeting. John sent a file.

Done Edit

Notification Understanding

Which characteristics best describe this notification?

The intelligent system currently considers this a **messaging & social** notification.

Professor 4 minutes ago
Check this paper.

☐ Messaging
☐ Social
☐ Mail
☐ Advertisements
☐ Gaming
☐ Others []

Is this notification a **to-do item**? (Swipe right if yes, left if not)

If a notification is a to-do item, the intelligent system will:

1. Rank it higher
2. Categorize as **To-Do**
3. Summarize the task

Professor 4 minutes ago
Check this paper.

abc_123456 1 hour ago
This restaurant seems great!

Which sentences best describe the nature of this notification?

David 2 hours ago
When's the meeting tomorrow?

☐ This notification is **urgent** because you often click on messages from David.

☐ This notification is **urgent** because it asks *When*.

☐ This notification is **work-related** because it mentions the term *meeting*.

☐ This notification is a **to-do item** because "?" means it needs a reply.

The system is unsure what this term means.

What does the highlighted text mean for this notification?

John Doe Yesterday
Project Documents ...

☐ It is work-related
☐ It involves interacting with others
☐ It is an advertisement
☐ It is a system notice
☐ Others []

Figure 1: Examples of Design Probes Shown to Interview Participants

process helped ensure the analysis remained grounded in the data while accommodating new insights.

Through this process, we identified a range of themes related to how users perceive and engage with human-in-the-loop notification systems. These themes form the foundation for the findings presented in the following sections.

3 Findings

3.1 Smartphone Users' Preferred Forms of Human-in-the-loop Interaction

3.1.1 Preference for Lightweight Interactions with Willingness to Express Nuanced Needs. Participants generally preferred lightweight HITL interactions to convey their notification preferences. Binary inputs, such as selecting *yes* or *no* to indicate whether a notification is important, should be prioritized, or belongs to a particular category, were commonly favored. These inputs required minimal physical effort and imposed low cognitive demand, allowing participants to express preferences without the need to make complex decisions.

However, when binary inputs were insufficient to represent more nuanced or conditional preferences, participants were willing to engage in more effortful interactions. These included multiple selections, rating scales, drag-and-drop sorting, and free-text or dialogue-based input. For example, P06 noted a willingness to define specific keywords that describe what qualifies as a *"must-not-miss"* notification, employing custom inputs to precisely determine how notifications should be handled.

3.1.2 Diverse Preferences for Expressing Needs Across Functions in a Notification System. Participants demonstrated differing preferences for how they wished to express their needs depending on the specific function of the notification system, such as categorization, summarization, and sorting.

For categorization, most participants preferred assigning a clear category to each notification rather than expressing preferences through rating scales or comparative judgments. P13 disliked the cognitive effort involved in providing nuanced responses for this task, stating: *"I'd rather just put it directly into the 'work' category without having to think about whether it's, say, 25% work or 100% work."*

When expressing summarization preferences, participants' approaches varied depending on their familiarity with the content. If they were able to anticipate the content based on past experience and had a clear idea of what information they needed, they preferred having more control over the summary. P05, for instance, described work-related messages as typically containing explicit instructions and preferred to specify key elements to include in the summary: *"Work-related messages are typically very explicit—they tell me exactly what to do."* In contrast, when dealing with notifications containing less predictable or more heterogeneous content, such as advertisements, participants preferred that the system first generate candidate summaries for them to select from.

Preferences for sorting notifications were more varied. Some participants favored expressing relative priority among notifications, while others preferred defining rules for prioritizing specific

types of content. Those who preferred relative priority often found binary labels such as *important* and *not important* insufficient, especially when multiple notifications were similarly important. As P24 explained: *“I feel like if I have a lot of important messages and I mark all of them as ‘important,’ then the system still won’t know how to order them properly.”* These concerns led some participants to favor drag-and-drop sorting as a more explicit way to indicate the order among sampled notifications.

At the same time, participants raised concerns about systems that rely solely on comparisons of sampled notifications. When all sampled items were considered important, participants worried that placing one lower in the list might signal to the system that it is less important overall. P11 illustrated this risk: *“If you use a sorting interface for work notifications, and everything is equally important, then by moving something lower in the list, you might end up overlooking it—even if it’s actually urgent—just because it’s near the bottom.”* To mitigate this risk, some participants preferred specifying characteristics of high-priority notifications, allowing the system to prioritize entire groups and reduce the chance of overlooking individual cases.

3.2 Smartphone Users’ Preferred Timing of Human-in-the-loop Interaction

3.2.1 Attentional and Physical Constraints Shape the Timing of HITL Interactions. Participants’ willingness to engage in HITL interactions was shaped by a combination of temporal, physical, and cognitive constraints encountered during daily smartphone use. Time was the most frequently cited constraint. When only brief moments were available—such as “between tasks at work” (P32)—participants favored interactions that were quick, lightweight, and minimally disruptive.

Physical conditions also limited engagement, especially for interactions that required precise or prolonged gestures. For instance, P30 noted that drag-and-drop operations were infeasible while standing on moving transportation. In such contexts, she expressed a preference for interactions that could be completed with a single click. Similarly, cognitive load reduced users’ ability to articulate preferences. Several participants (P05, P30, P31) described difficulty in setting preferences when they were already focused on tasks like working or commuting. In these contexts, participants favored simpler and more intuitive interaction mechanisms, such as binary options for prioritizing notifications, which required minimal cognitive investment.

Emotional and energy states further affected engagement. Fatigue was commonly mentioned as a deterrent. As P22 reflected, *“I’m already exhausted from commuting to and from work—I just don’t want to deal with it.”*

3.2.2 Aligning HITL Timing with Notification-Handling Routines. Participants’ willingness to engage in HITL interactions was shaped not only by the above constraints, but also by how well these interactions aligned with their existing notification-handling routines. In particular, users expected HITL prompts to respect their temporal preferences for when they typically attended to notifications. Interactions occurring outside these periods were seen as disruptive. As P32 elaborated, *“The reason I initially divided the day into specific time slots for checking notifications was to ensure I could focus on my*

tasks in between those periods. Therefore, if I am to interact with the notification system, I expect such [HITL] interactions to occur strictly within those designated time slots.”

However, participants also made a finer distinction: they did not welcome HITL prompts at just any point within a notification session. Rather, they expressed a clear preference for being prompted only after completing a notification-related task, such as reading or processing a message. Prompts that appeared mid-task were experienced as intrusive and attentionally disruptive. For example, P10 noted, *“It would be less disruptive if the prompt appeared only after I had finished reading and processing the message. If it pops up while I’m still reading, it interrupts my attention. Ideally, the system should wait until I’m about to exit or have clearly completed the reading before initiating the prompt.”*

On the other hand, participants selectively avoided HITL interactions for notifications they found emotionally taxing or irrelevant to the moment. Because they typically disliked encountering these notifications during certain times, they were also unwilling to answer questions related to them, as doing so would require revisiting unwanted content. For example, participants expressed reluctance to engage with HITL interactions about work-related notifications during off-hours or rest periods. P30 remarked, *“If it’s about work, I don’t want to deal with it when I’ve just woken up—let alone before bed. I already have trouble sleeping, and I’m afraid one keyword will trigger something and I won’t sleep all night.”* Conversely, asking about leisure content could feel disruptive during periods of focused work. P33 illustrated this clearly: *“I don’t want ads drawing my attention away at work. If it’s something like a Facebook message, I might get curious about how it’s prioritized, [...] but the way you think while reading ad-related content is totally different from how you think during work—it just breaks your rhythm.”*

These insights highlight the importance of designing HITL timing mechanisms that go beyond assumptions about user availability or idle moments. Effective timing should be grounded in users’ existing notification interaction routines—specifically, times when they are already cognitively and emotionally prepared to engage, and when the content aligns with their current context.

4 Conclusion and Future Work

This study provides an initial understanding of how participants prefer HITL interactions to be integrated into intelligent smartphone notification systems for personalization. Low-effort mechanisms were generally preferred, but participants were willing to engage in more effortful interactions when these enabled them to express nuanced preferences that better reflected their intentions. HITL interactions were more acceptable when they aligned with participants’ existing notification-handling routines and matched their cognitive and physical availability. We will conduct a deeper analysis of the interview data to examine the factors that influence participants’ willingness to engage in HITL interactions, how these factors interact, and the expectations or concerns users hold toward notification systems that incorporate HITL features.

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